

```
experiment3.l X
EXP3 > experiment3.l
1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  int|float|char  {
8  | printf("Keyword : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]*  {
12  | printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+  {
16  | printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+  {
20  | /* Ignore white spaces */
21  }
22
23  .  {
24  | printf("Unknown Token : %s\n", yytext);
25  }
26
27  %%
28
29  int main()
30  {
31  | printf("Enter Source Code:\n\n");
32  | yylex();
33  | return 0;
34  }
35
36  int yywrap()
37  {
38  | return 1;
39  }
```

```
M /c/Users/Afroz/Desktop/PCD/EXP3
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ls -l scanner.exe
-rwxr-xr-x 1 Afroz Afroz 128058 Aug 10 20:50 scanner.exe
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ./scanner.exe
Enter Source Code:
int total = 100;
Keyword : int
Identifier : total
Unknown Token : =
Integer : 100
Unknown Token : ;
```

EXP3 > experiment3_activity1.l

```
1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  ∨ [a-zA-Z][a-zA-Z0-9_]* {
8  |   printf("Identifier : %s\n", yytext);
9  | }
10
11 ∨ [0-9]+ {
12 |   printf("Integer : %s\n", yytext);
13 | }
14
15 ∨ [ \t\n]+ {
16 |   /* Ignore white spaces */
17 | }
18
19 ∨ "+"|"-"|"*"|" "/"|"%" {
20 |   printf("Arithmetic Operator : %s\n", yytext);
21 | }
22
23 ∨ . {
24 |   printf("Unknown Token : %s\n", yytext);
25 | }
26
27 %%
28
29 int main()
30 ∨ {
31 |   printf("Enter Source Code:\n\n");
32 |   yylex();
33 |   return 0;
34 | }
35
36 int yywrap()
37 ∨ {
38 |   return 1;
39 | }
```

 /c/Users/Afroz/Desktop/PCD/EXP3

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
\$ flex experiment3_activity1.l

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
\$ gcc lex.yy.c -o activity1

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
\$./activity1.exe
Enter Source Code:

a + b * c - 20 / 5
Identifier : a
Arithmetic Operator : +
Identifier : b
Arithmetic Operator : *
Identifier : c
Arithmetic Operator : -
Integer : 20
Arithmetic Operator : /
Integer : 5

EXP3 > experiment3_activity2.l

```
1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  int|float|char    {
8      printf("Keyword : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]*    {
12      printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+    {
16      printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+    {
20      /* Ignore white spaces */
21  }
22
23  ";"|","|"(")"|"{"}"    {
24      printf("Delimiter : %s\n", yytext);
25  }
26
27  .    {
28      printf("Unknown Token : %s\n", yytext);
29  }
30
31  %%
32
33  int main()
34  {
35      printf("Enter Source Code:\n\n");
36      yylex();
37      return 0;
38  }
39
40  int yywrap()
41  {
42      return 1;
43  }
```

```
M /c/Users/Afroz/Desktop/PCD/EXP3
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ flex experiment3_activity2.l
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ gcc lex.yy.c -o activity2
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ./activity2.exe
Enter Source Code:

int main()
Keyword : int
Identifier : main
Delimiter : (
Delimiter : )
{
Delimiter : {
}
Delimiter : }
```

```

1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  int|float|char    {
8      printf("Keyword : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]*    {
12      printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+    {
16      printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+    {
20      /* Ignore white spaces */
21  }
22
23  ";"|","|"(")|"|"{"|"}"    {
24      printf("Delimiter : %s\n", yytext);
25  }
26
27  .    {
28      printf("Unknown Token : %s\n", yytext);
29  }
30
31  %%
32
33  int main()
34  {
35      printf("Enter Source Code:\n\n");
36      yylex();
37      return 0;
38  }
39
40  int yywrap()
41  {
42      return 1;
43  }

```

```

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ flex experiment3_activity3.1

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ gcc lex.yy.c -o activity3

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ./activity3.exe
Enter Source Code:

int age = 20;
Keyword : int
Identifier : age
Unknown Token : =
Integer : 20
Delimiter : ;
float salary = 25000;
Keyword : float
Identifier : salary
Unknown Token : =
Integer : 25000
Delimiter : ;
char grade;
Keyword : char
Identifier : grade
Delimiter : ;

```

EXP3 > experiment3_student.l

```

1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  int|float|char|if|else|while|for|return    {
8      printf("Keyword : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]*    {
12      printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+    {
16      printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+    {
20      /* Ignore white spaces */
21  }
22
23  ";"|","|"(")"|"{"|"}"    {
24      printf("Delimiter : %s\n", yytext);
25  }
26
27  .    {
28      printf("Unknown Token : %s\n", yytext);
29  }
30
31  %%
32
33  int main()
34  {
35      printf("Enter Source Code:\n\n");
36      yylex();
37      return 0;
38  }
39
40  int yywrap()
41  {
42      return 1;
43  }

```

```

experiment3_additional.l
EXP3 > experiment3_additional.l
1  %{
2  #include <stdio.h>
3  %{
4
5  %%
6
7  "=="|"!="|"<="|">="|"<"|>" {
8  |   printf("Relational Operator : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]* {
12  |   printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+ {
16  |   printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+ {
20  |   /* Ignore white spaces */
21  }
22
23  . {
24  |   printf("Unknown Token : %s\n", yytext);
25  }
26
27  %%
28
29  int main()
30  {
31  |   printf("Enter Source Code:\n\n");
32  |   yylex();
33  |   return 0;
34  }
35
36  int yywrap()
37  {
38  |   return 1;
39  }

```

```

M /c/Users/Afroz/Desktop/PCD/EXP3
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ flex experiment3_student.l
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ gcc lex.yy.c -o student
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ./student.exe
Enter Source Code:
if (age > 18)
Keyword : if
Delimiter : (
Identifier : age
Unknown Token : >
Integer : 18
Delimiter : )
{
Delimiter : {
return age;
Keyword : return
Identifier : age
Delimiter : ;
}
Delimiter : }
else
Keyword : else
{
Delimiter : {
return 0;
Keyword : return
Integer : 0
Delimiter : ;
}
Delimiter : }

```

experiment3_additional.l

EXP3 > experiment3_additional.l

```
1  %{
2  #include <stdio.h>
3  %}
4
5  %%
6
7  "=="|"!="|"<="|">="|"<"|>" {
8      printf("Relational Operator : %s\n", yytext);
9  }
10
11  [a-zA-Z_][a-zA-Z0-9_]* {
12      printf("Identifier : %s\n", yytext);
13  }
14
15  [0-9]+ {
16      printf("Integer : %s\n", yytext);
17  }
18
19  [ \t\n]+ {
20      /* Ignore white spaces */
21  }
22
23  . {
24      printf("Unknown Token : %s\n", yytext);
25  }
26
27  %%
28
29  int main()
30  {
31      printf("Enter Source Code:\n\n");
32      yylex();
33      return 0;
34  }
35
36  int yywrap()
37  {
38      return 1;
39  }
```

```
/c/Users/Afroz/Desktop/PCD/EXP3
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ flex experiment3_additional.l
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ gcc lex.yy.c -o additional
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/PCD/EXP3
$ ./additional.exe
Enter Source Code:
a == b
Identifier : a
Relational Operator : ==
Identifier : b
x != y
Identifier : x
Relational Operator : !=
Identifier : y
p < q
Identifier : p
Relational Operator : <
Identifier : q
m > n
Identifier : m
Relational Operator : >
Identifier : n
a <= b
Identifier : a
Relational Operator : <=
Identifier : b
c >= d
Identifier : c
Relational Operator : >=
Identifier : d
```